

[Training Data]

All verb/object primitives are learned

	o_1	o_2	o_3	o_4	o_5
v_1	Train Pairs	Train Pairs	Train Pairs	Train Pairs	Train Pairs
v_2	Train Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs
v_3	Train Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs
v_4	Train Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs
v_5	Train Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs

Train Pairs

[Closed-world Evaluation]

Evaluate restricted to the pairs present in the dataset

	o_1	o_2	o_3	o_4	o_5
v_1	Train Pairs	Test Seen Pairs	Test Seen Pairs	Train Pairs	Train Pairs
v_2	Train Pairs	Masked Pairs (Ignored)	Test Unseen Pairs	Test Unseen Pairs	Masked Pairs (Ignored)
v_3	Test Seen Pairs	Masked Pairs (Ignored)	Test Unseen Pairs	Masked Pairs (Ignored)	Masked Pairs (Ignored)
v_4	Test Seen Pairs	Test Unseen Pairs	Test Unseen Pairs	Test Unseen Pairs	Masked Pairs (Ignored)
v_5	Train Pairs	Masked Pairs (Ignored)	Masked Pairs (Ignored)	Masked Pairs (Ignored)	Masked Pairs (Ignored)

Test Seen Pairs

Test Unseen Pairs

Masked Pairs (Ignored)

[Open-world Evaluation]

Evaluate across the entire space

	o_1	o_2	o_3	o_4	o_5
v_1	Train Pairs	Test Seen Pairs	Test Seen Pairs	Train Pairs	Train Pairs
v_2	Train Pairs	Possible Unseen Pairs	Test Unseen Pairs	Test Unseen Pairs	Possible Unseen Pairs
v_3	Test Seen Pairs	Possible Unseen Pairs	Test Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs
v_4	Test Seen Pairs	Test Unseen Pairs	Test Unseen Pairs	Test Unseen Pairs	Possible Unseen Pairs
v_5	Train Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs	Possible Unseen Pairs

Test Seen Pairs

Test Unseen Pairs

Possible Unseen Pairs